

FABRICATION & INSTALLATION GUIDE

J40 Integrated Cooling Pack

Engine radiator • Turbo intercooler • A/C condenser • Receiver-drier • Fans

REV B | 23 JULY 2026 | MOCK-UP AND QUOTATION RELEASE

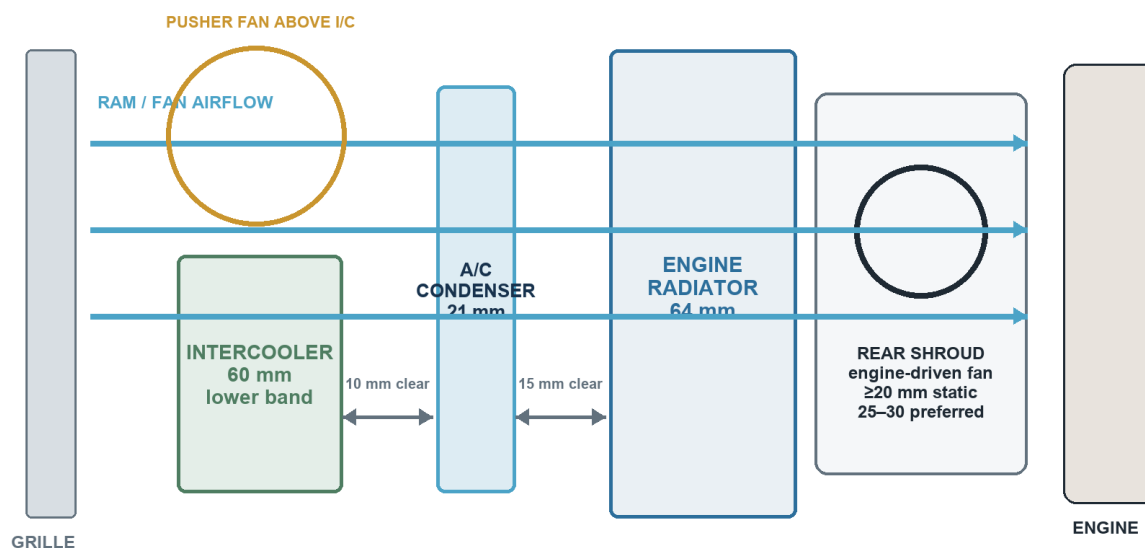
STATUS	STRUCTURAL DECISION	FINAL-CORE RELEASE
Mock-up approved	Retain existing upright; weld a mirrored upright on the left	HOLD until M1–M7 are recorded

DESIGN INTENT

Build one modular front cooling package. Keep all pressure circuits separate and every component individually removable. No exchanger carries another exchanger's weight.

SIDE SECTION — AIRFLOW AND REQUIRED GAPS

Nominal lower-band stack depth: 170 mm; allow 180–190 mm installed.

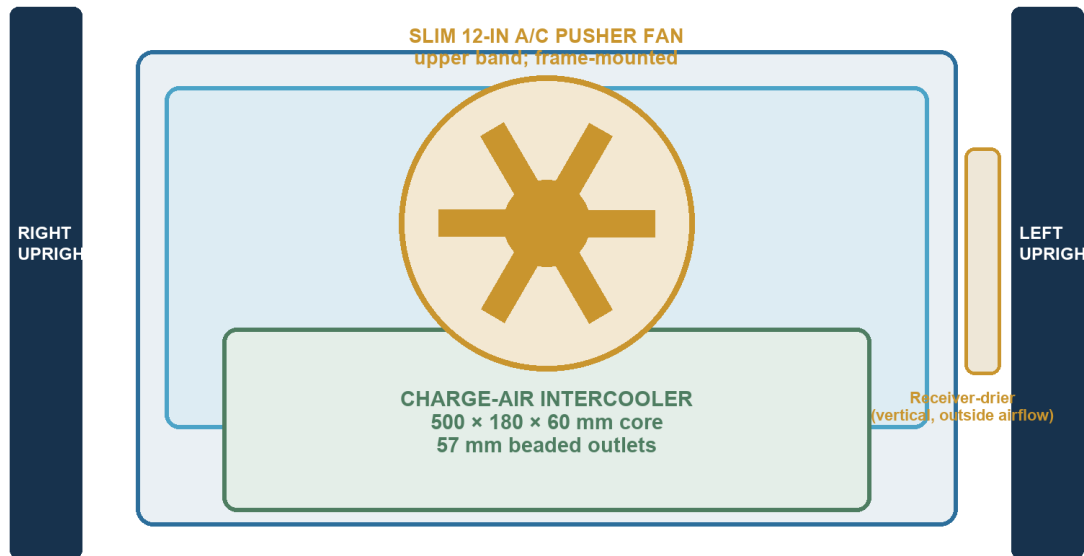


1. Assembly at a glance

The front pack has three heat-exchanger cores and two nearby A/C/fan auxiliaries. The receiver-drier is not a fourth radiator. The turbo itself remains on the engine; only its charge-air intercooler is in the front stack.

FRONT ELEVATION — LOOKING THROUGH THE GRILLE

Component bodies shown; mounting tabs and tank seams remain site-fit.



What must be independently removable

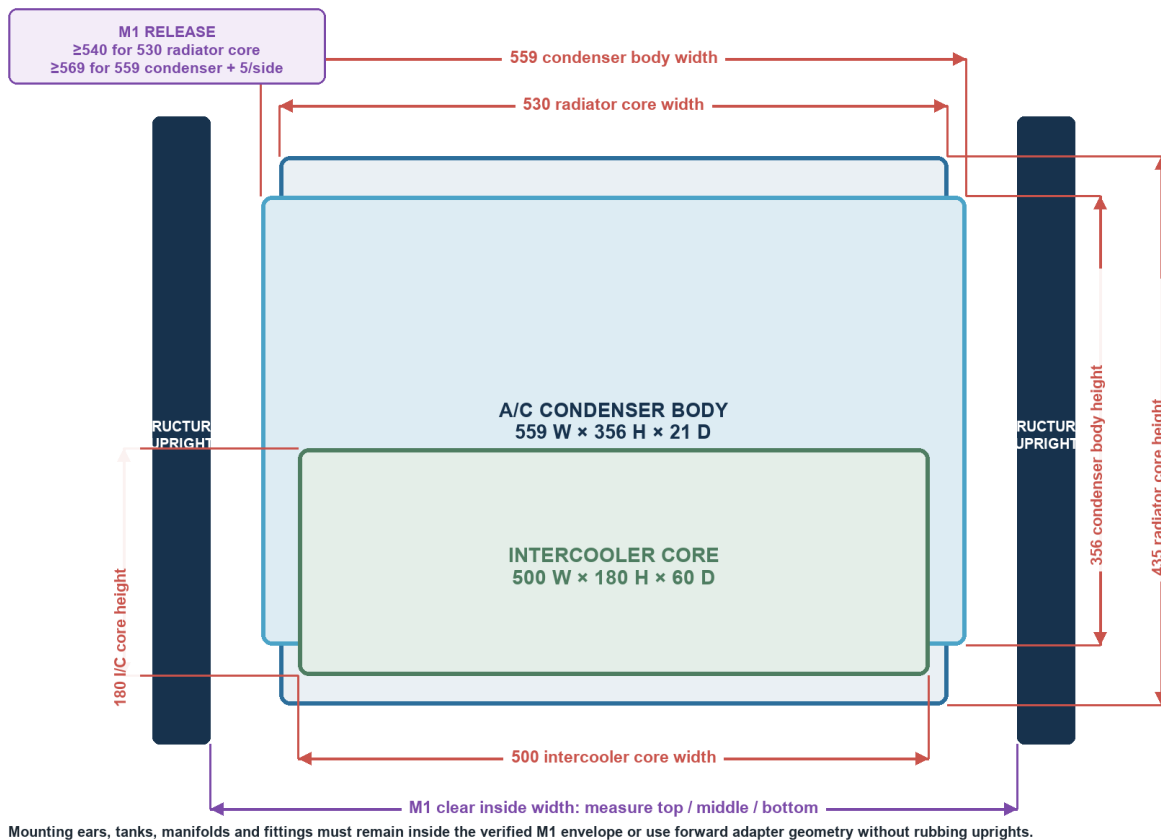
- Engine radiator, rear shroud and their rubber-isolated saddles/tabs.
- A/C condenser, receiver-drier and slim pusher fan.
- Turbo charge-air intercooler and its four isolated mounting tabs.
- Side rails and any crossrails needed for component removal or service.

2. Dimensioned fabrication sheets

Dimension authority: red dimensions are the Rev B manufacture targets. Purple M-dimensions are mandatory vehicle measurements and override assumptions. Do not scale the printed drawing; use the written dimensions.

DIMENSIONED FRONT ELEVATION — CORE BODIES

All values mm. Red = manufacture target. Purple M-dimensions = measure on vehicle before release.

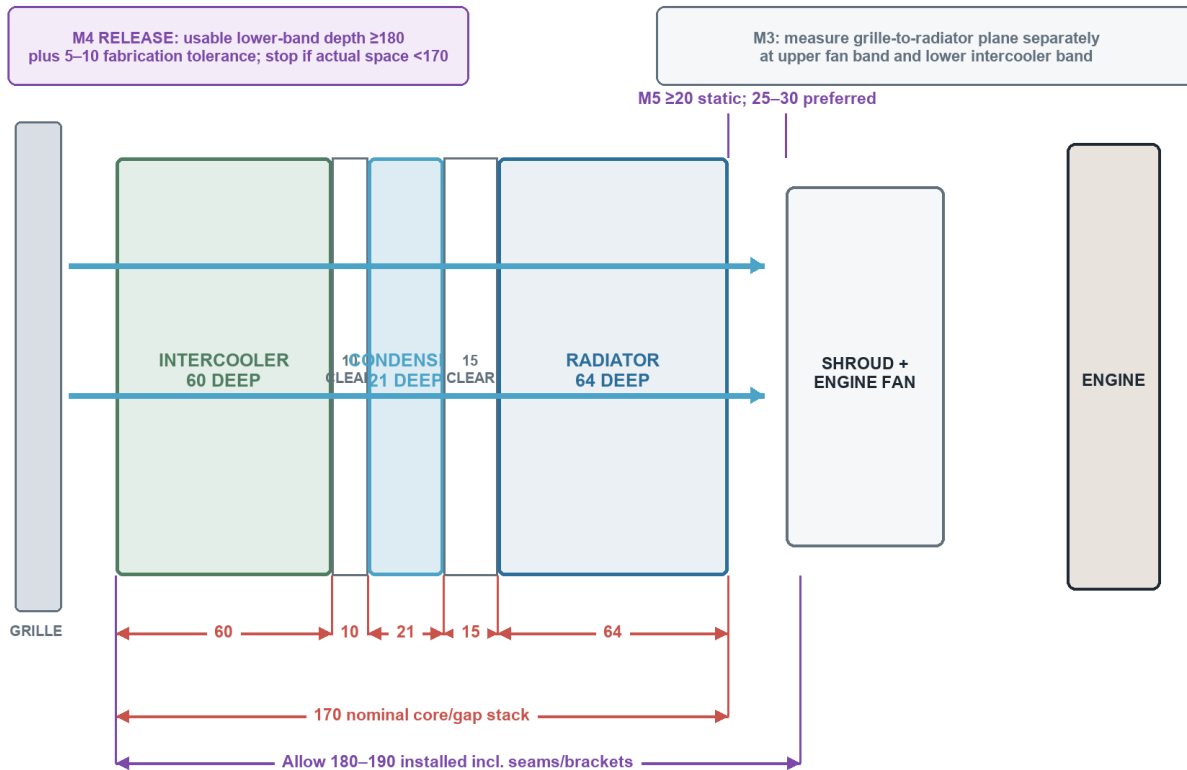


FRONT-ELEVATION RELEASE	REQUIREMENT
Radiator	530 $\times$ 435 core; overall width $\leq$ M1 – 10
Condenser	559 $\times$ 356 body nominal; verify M1 ≥ 569 or use proven forward adapter geometry
Intercooler	500 $\times$ 180 core; overall width $\leq$ M1 – 10
Vertical placement	Site-fit from M2, M6, bonnet/latch and service-removal checks

2A. Dimensioned side section

DIMENSIONED SIDE SECTION — LOWER AIRFLOW BAND

All values mm. Component thicknesses + clear gaps form the controlling front-to-rear stack.



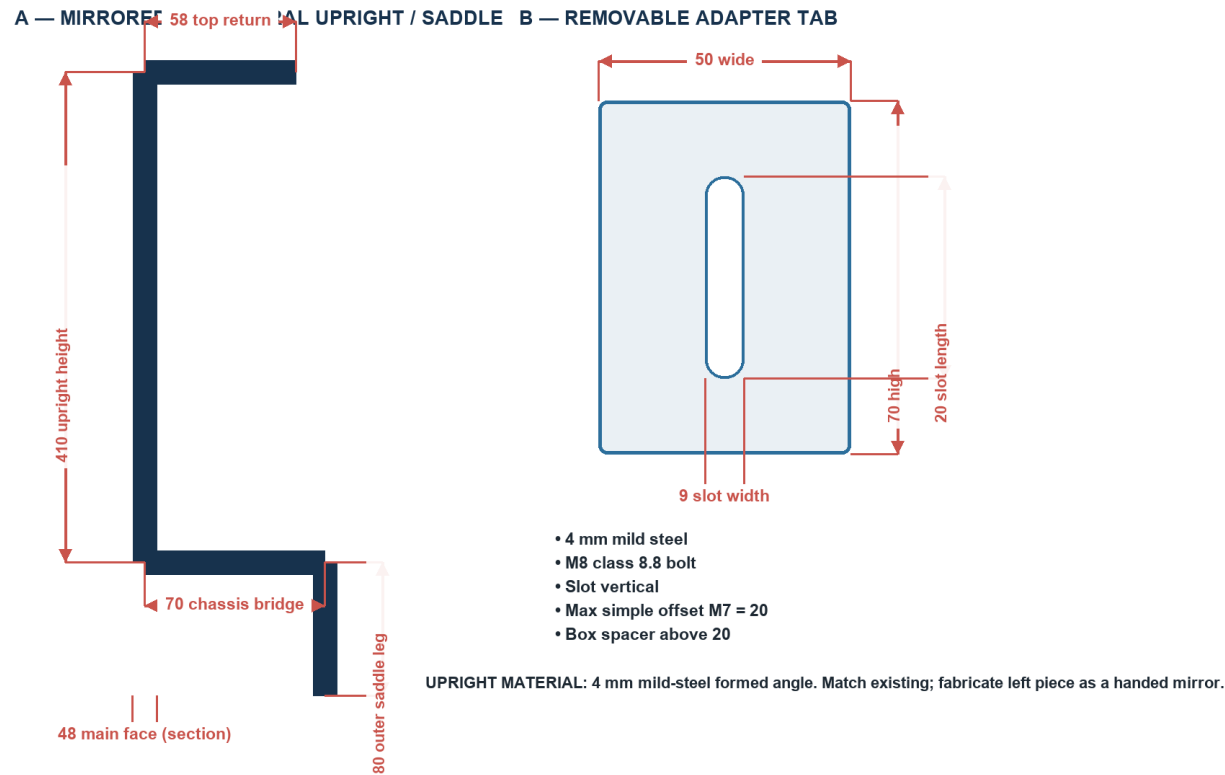
Controlling arithmetic: 60 intercooler + 10 clear + 21 condenser + 15 clear + 64 radiator = 170 mm nominal.

Allow 180–190 mm for seams, brackets and imperfect planes. The fan and shroud sit behind the radiator and are controlled separately by M5.

2B. Dimensioned mounting details

DIMENSIONED MOUNTING DETAILS — REV B

All values mm. Verify handed geometry on the vehicle before cutting the mirrored upright.



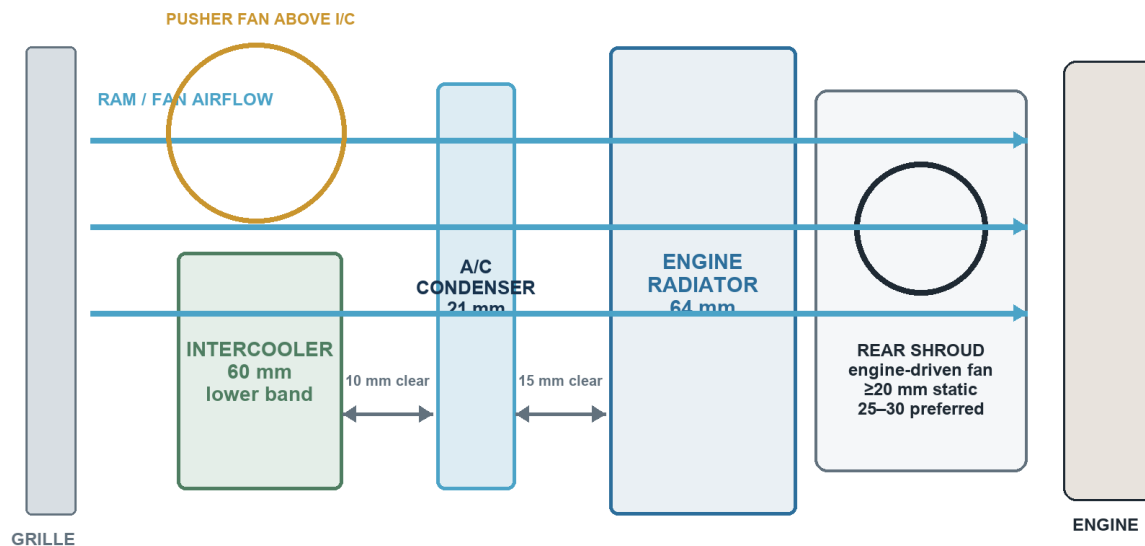
DETAIL	DIMENSION / MATERIAL
Removable side rail	30 × 3 flat OR 25 × 25 × 3 angle; cut length M2 – 10; keep 5 top and bottom clearance
Radiator lower saddle	3–4 plate with 5 EPDM pad; upper tabs restrain only
Condenser tabs	3 plate, rubber washers and M6 bolts
Intercooler tabs	4 plate, M8 isolators; 2–3 thermal-movement slot at one upper point

The upright dimensions reproduce the measured existing-side bracket basis. Mirror and verify the left part against the chassis before welding. Slots belong in removable adapter tabs, never in structural uprights.

3. Front-to-rear package

SIDE SECTION — AIRFLOW AND REQUIRED GAPS

Nominal lower-band stack depth: 170 mm; allow 180–190 mm installed.



ORDER	COMPONENT	DESIGN ENVELOPE / RULE
1 — front upper	Slim A/C pusher fan	12 in nominal; ≤330 mm diameter × 65 mm deep; upper band
1 — front lower	Charge-air intercooler	500 W × 180 H × 60 D mm; 57 mm OD outlets
2	Clear air gap	10 mm minimum between intercooler and condenser
3	R134a condenser	559 W × 356 H × 21 D mm nominal; ≤600 × 370 × 25
4	Clear air gap	15 mm target; 10 mm absolute minimum to radiator
5 — rear	Engine radiator	530 W × 435 H × 64 D mm core; high-efficiency four-row pattern
6 — engine side	Rear shroud and mechanical fan	≥20 mm static tip clearance; 25–30 mm preferred

4. Separate circuits and connections

THREE SEPARATE CIRCUITS — NEVER PLUMB THEM TOGETHER

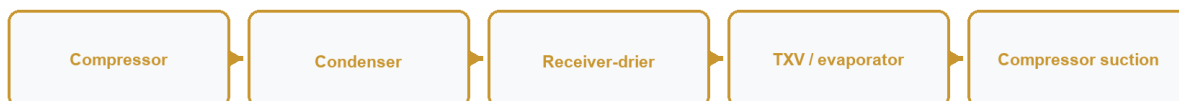
ENGINE COOLANT



TURBO CHARGE AIR



R134a AIR CONDITIONING



Receiver-drier is A/C hardware mounted beside the condenser—not a fourth heat-exchanger core.

Interface holds

- Copy radiator neck side, angle and centre position from the old radiator and installed hose mock-up; target neck OD is 38 mm.
- Do not finalize intercooler outlet direction until turbo compressor outlet and intake-plenum inlet are marked on the vehicle.
- Confirm condenser #8 inlet and #6 outlet threads before crimping hoses; mount the receiver-drier vertically after the condenser outlet.
- Do not open the new receiver-drier to atmosphere until final A/C assembly and charging preparation.

5. Structure and mounting

Fixed structural decision: retain the measured right-side formed upright and fabricate a handed/mirrored upright on the left. Dry-fit the complete pack before final welding or drilling.

ITEM	FABRICATION REQUIREMENT
Structural uprights	410 mm high; 48 mm main face; 4 mm mild steel; mirror existing 58 mm top return, 70 mm bridge and 80 mm saddle leg.
Removable side rails	30 × 3 mm flat or 25 × 25 × 3 mm angle; final length M2 – 10 mm.
Rail attachment	Two M8 class 8.8 bolts per upright; four total. Put 9 × 20 mm vertical slots in adapter tabs —not structural uprights.
Radiator	Weight on two 3–4 mm lower saddles with 5 mm EPDM pads. Upper tabs restrain only.
Condenser	Four independent 3 mm tabs, rubber washers and M6 bolts.
Intercooler	Four independent 4 mm tabs and M8 isolators; one upper point gets 2–3 mm horizontal thermal-movement slot.
Fan	Removable hoop/crossrail. Never use through-core plastic rods.
Receiver-drier	Rubber-lined removable vertical clamp outside the primary airflow.

Absolute no-go conditions

- No welding, drilling or hard clamping through tanks, tubes, cores or fins.
- No radiator carrying the condenser, intercooler or fan.
- No condenser carrying the fan or intercooler.
- No forced bolts that twist a core or pull the radiator toward the engine fan.
- No permanent crossrail across the radiator core face.

6. Mandatory vehicle measurements — M1 to M7

Record these with the second upright tacked in place and the grille/front panel, bonnet latch, engine fan and body position represented. Final core manufacture remains on hold until all seven are complete.

ID	MEASUREMENT	METHOD	PASS / RELEASE CRITERION
M1	Minimum clear inside width	Top / middle / bottom; use smallest	≥540 for 530 core; ≥569 for unmodified 559 condenser + clearance
M2	Clear vertical opening	Lower pad plane to bonnet/latch obstruction	Radiator overall height + 10 mm
M3	Grille to radiator front plane	Upper fan band and lower I/C band	Must accept chosen front components
M4	Usable lower-band stack depth	Grille/guard to radiator rear-component datum	≥180 mm plus 5–10 mm tolerance
M5	Radiator rear face to fan tips	Closest blade through full rotation	≥20 mm; 25–30 preferred
M6	Lowest safe intercooler edge	Relative to protected frame/bumper line	≥25 mm above lowest protected line
M7	Upright face to side-rail offset	All four corners	0–20 simple tab; boxed spacer above 20

Also capture

- Coolant-neck centres, OD and angles; cap, overflow and drain positions.
- Condenser port side and actual threads.
- Turbo compressor outlet and intake-plenum inlet locations.
- Bonnet closure, steering/body clearance and service-removal path.
- A ruler/tape visible in every measurement photograph.

7. Fabricator sequence

1. Duplicate the existing upright as a left-hand mirror; tack it and remeasure the opening.
2. Make full-size radiator, condenser, intercooler and fan envelopes from cardboard or plywood.
3. Set the radiator plane from rear fan clearance and coolant-hose sweep.
4. Position the condenser with its clear air gap, then place the intercooler only in the lower frontal band.
5. Place the pusher fan principally in the upper band and verify grille/bonnet-latch clearance.
6. Mark removable rails, tabs, saddles and the vertical receiver-drier clamp from the real components.
7. Confirm charge-pipe and A/C hose routes before outlet welding or hose crimping.
8. Record M1–M7, photograph the mock-up, and obtain owner release for final cores.
9. Bench pressure/flow-test the radiator; leak-test the intercooler at 20 psi; obtain A/C-shop condenser evidence.
10. Final-fit, corrosion-protect, install and validate at warm idle with A/C on.

Decision rules if depth is short

- Protect radiator size and rear fan clearance first.
- Reduce intercooler thickness from 60 to 50 mm only with a reputable high-efficiency core.
- Move or split the pusher fan before reducing radiator area.
- If lower-band usable depth is under 170 mm, stop and review a remote/water-to-air charge cooler.

8. Inspection and handover record

CHECK	ACCEPTANCE ITEM
☐	M1–M7 sheet complete with ruler photographs
☐	Front and side full-size mock-up photographs
☐	Grille, bonnet/latch and fan represented during tack-fit
☐	As-built sketch: cores, tanks, ports, holes and offsets
☐	Radiator pressure and flow result recorded
☐	Intercooler 20 psi leak-test result recorded
☐	Condenser/A/C pressure-test evidence recorded
☐	Bare-metal bracket and weld photographs
☐	No core carries another component
☐	Fan airflow direction confirmed with paper-strip test
☐	Warm-idle and A/C-on validation complete
☐	All exchangers removable without cutting welded parts

Release note

This document releases mock-up, quotation and measurement capture. It does not release final core manufacture until M1–M7 and the actual component interfaces have been approved.

Controlled project references

- docs/j40-integrated-cooling-pack-fabricator-handoff-20260717.md
- data/manual/fabrication/front_cooling_stack_rev_a/integrated_cooling_pack_dimensions_rev_b.csv
- docs/engine-radiator-recore-release-20260529.md
- docs/2h-turbo-suitability-and-options-20260717.md
- docs/ac-hvac-workstream.md